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Optimized Ensemble Learning with SMOTE-Tomek Resampling for Early Detection of Hepatitis B

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***Abstract*— Hepatitis B is a major worldwide health issue because, if left untreated, it can cause serious liver problems and a high death rate. Despite the availability of treatments, early detection and accurate projections are still difficult to achieve due to imbalanced data, missing clinical values, and diagnosis difficulties. The goal of this paper is to develop a reliable and clinically relevant prediction model for Hepatitis B by using sophisticated machine learning methods. A robust preprocessing pipeline, including KNN-based missing value imputation, label encoding, and SMOTE-Tomek hybrid resampling, was applied to address class imbalance and guarantee data quality. Seven machine learning algorithms including Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), Logistic Regression, Naive Bayes, Decision Tree and XGBoost were assessed utilizing a publicly accessible dataset. Furthermore, a stacked ensemble model combining SVM, XGBoost and tuned Random Forest was proposed to enhance the overall quality of predictions by using the advantages of distinct models. The Stacked Ensemble outperformed every individual model, achieving the maximum accuracy of 97.9%, precision, and F1-score of nearly 99% and 95.6%, respectively. The findings highlight how hybrid resampling and optimized ensemble learning can significantly enhance the predictive accuracy of Hepatitis B outcome models.**

Keywords—Hepatitis B, Machine Learning, Stacked Ensemble, SMOTE-Tomek, Oversampling.

I. Introduction

Approximately two hundred million individuals globally are afflicted by the hepatitis B virus (HBV), which continues to be ranked among the most dangerous along with pervasive worldwide illnesses [1]. Moreover, three hundred million individuals are still ongoing carriers, with roughly 75 percent of them living in the region that includes Asia and the Pacific, even though highly successful medicines have been available from 1982 [2]. Hepatocellular carcinoma (HCC), cirrhosis, and hepatic dysfunction are all markedly increased by continuous infection with HBV [3]. Understanding the cellular mechanisms of infections caused by HBV seems essential to bolstering present attempts to create treatments [4]. Entecavir (ETV) is a potent nucleoside analogue (NA) that is frequently used for treating Hepatitis and is among the more successful NAs [5]. There is minimum 8 genotypes of HBV (A–H), along with every has unique medical features plus regional patterns. Genotypes A and D are most common in Africa, Europe, and India, whereas genotypes B and C predominate in Asia [6]. Compared with mothers with antibody to e, HBeAg-positive mothers are very likely to transmit HBV to their babies [7]. Expanding vaccination will be necessary to meet worldwide reduction goals, which are outlined in the WHO Global Health Sector Strategy (GHSS), but it also requires proactively case detection and better chances for medication, both of which will be aided through a better biological knowledge of HBV-host connections [8].

- **Comprehensive Data Preprocessing:** Missing values and class imbalance were handles using KNN imputation, label encoding & SMOTE-Tomek resampling.
- **Extensive Algorithm Evaluation:** Seven machine learning algorithms were trained & evaluated using standard evaluation measures.
- **Superior Performance:** A stacked ensemble combining tuned random forest, SVM & XGBoost outperformed individual models.
- **Detailed Model Evaluation:** Performance was evaluated using accuracy, precision, recall, F1 score, ROC curve, and confusion matrix.

II. Literature Review

In order to measure the community-centered prevalence of hepatitis B virus, a random effects model centered around proportions was applied in an in-depth analysis of the research to collect main articles published between January 1, 2010, and December 31, 2019. In their study, a combined prevalence of 9.5% was found in 47 studies involving 21,702 respondents that were analyzed in the final analysis [9]. A study employed SHapley Additive exPlanations (SHAP), an interactive games based conceptual method was utilized in this research. The aim was to enhance transparency in model decision-making. The models achieved sequentially accuracies of 75%, 82%, 75%, 86%, 92%, and 90% [10]. Iannacone and Guidotti reviewed the immunobiology and pathogenesis of Hepatitis B virus infection, paying special attention to the function of CD8+ T cells and a number of recent discoveries that

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contradict accepted wisdom. But the study is mostly theoretical without actual encouragement; neither experimental research nor execution are present [11].

Through an analysis of the literature on OBI virology and epidemiology, the authors explore its potential for oncogenesis, transmission, and broader clinical consequences. According to their research, OBI can remain undiagnosed and cause active hepatitis while immunosuppressed, advance cirrhosis be spread by blood transfusions or organ transplants and maintain pro-oncogenic qualities, which raises the risk of hepatocellular carcinoma [12]. The study's authors examined important therapy turning points, focusing on interferon and antivirals. Since 1998, results have improved dramatically because of the development of strong nucleos(t)ide analogues, which slow the progression of the disease, reduce liver inflammation, and decrease viral multiplication [13]. In this research, scientists use the Virology-Informed Neural Networks (VINN) technique on a newly suggested hepatitis B virus (HBV) infection dynamic model to more precisely forecast how the virus will spread throughout the liver. This model's 4 sections include free viruses, rcDNA-containing capsids, infected and uninfected hepatocytes, and capsid recycling factor [14].

III. Materials & Method

Fig. 1 illustrates the entire process diagram for the suggested approach for detecting Hepatitis B virus. The process begins with comprehensive data preprocessing, which includes handling missing values using KNN imputer and converting categorical features into numerical format using Label encoder. To maintain medical significance, continuous variables were binned into clinically relevant ranges. After preprocessing, to identify the most relevant predictors, feature selection was carried out by combining the Chi-Square test with XGBoost feature importance. By creating synthetic minority samples & eliminating overlapping instances, a SMOTE- Tomek hybrid resampling technique was used to alleviate class imbalance. After that, it is split into training and testing sets: 20% for testing and 80% for training. Following that, a few of machine learning methods are trained and assessed, such as K-Nearest Neighbors (KNN), Random Forest, Naive Bayes, Decision Trees, Support Vector Machines (SVM), Logistic Regression & XGBoost. Furthermore, a Stacked Ensemble model was suggested, which used Logistic Regression as the meta-classifier and adjusted Random Forest, SVM, and XGBoost as base learners. Every model's performance is evaluated using the ROC curve, F1-score, confusion matrix, accuracy, precision, and recall. The entire process shown in Fig 1, from preprocessing to evaluation, is intended to select the model with the highest performance. Among the tested models, the proposed Stacked model delivered the most accurate result, achieving 97.9% accuracy in classification.

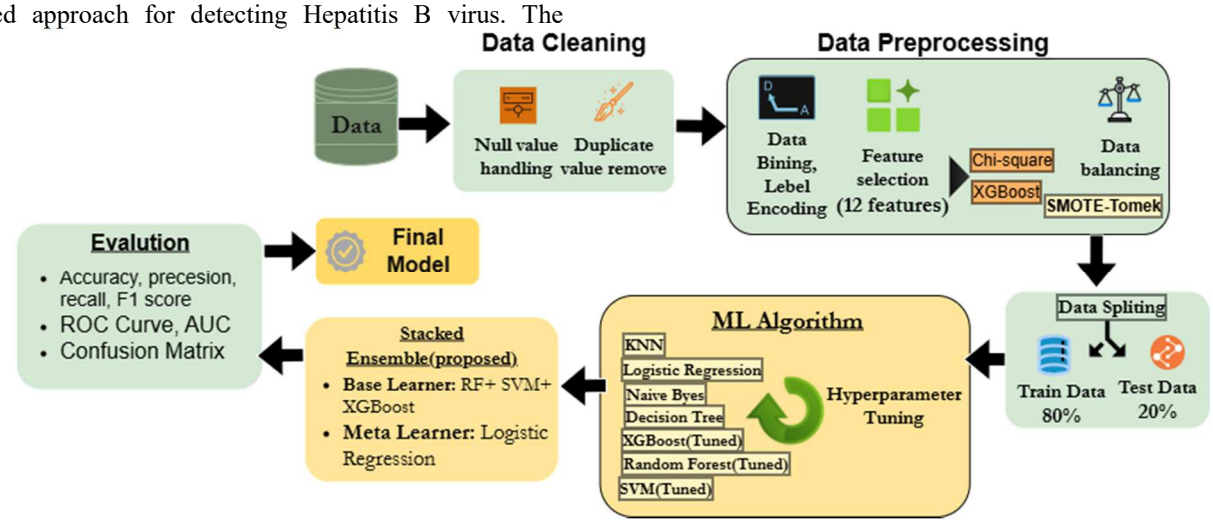


Fig. 1: The visual representation of proposed method

A. Dataset

The dataset used in this study is sourced from UCI Machine Learning Respiratory and originates from a clinical study of Hepatitis B patient's outcomes [15]. The dataset contains the medical profiles of 155 individuals with 20 features which includes demographic data (AGE, SEX), laboratory tests (SGOT, Albumin), clinical data (Fatigue, Malaise) and histological & physical findings (Histology results, Liver firmness).

In this study, 'Class' column is our target feature which indicates the survival status of a patient. According to the analysis, the majority of those infected were between the ages of 30 and 50, with an average age of roughly (Fig. 2). This implies that the most impacted demographic in this dataset is adults in their midlife.

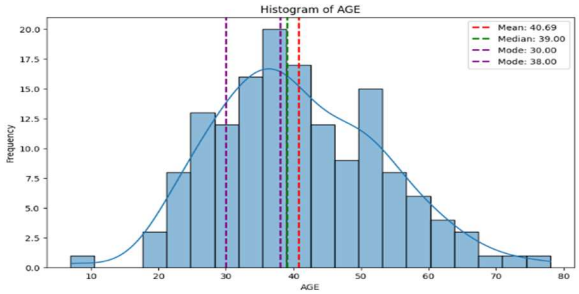


Fig. 2: Age-related risk of lung cancer

B. Data Pre-processing & Cleaning

Data Cleaning: Initially, there were 155 patients with 20 features in the original dataset. After the examination of the dataset, we had found a significant number of missing values

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in 2 columns (ALK PHOSPHATE & PROTIME). To protect the data integrity, those 2 columns were dropped from the dataset. In some columns, the dataset still had several missing values (Categorical feature like STEROID, LIVER BIG, LIVER FIRM etc & Numerical features like BILIRUBIN, SGOT, and ALBUMIN). The remaining missing values were handled using a KNN Imputer which estimates missing entries based on the similarity of other samples. After that, the dataset is verified to contain no null values. An inspection had also done to the dataset in order to find duplicate entries. If any duplicate rows found, we reviewed and removed it to avoid repetition.

Encoding: The dataset contains several categorical variables. To convert them in a format suitable for machine learning, we have used label encoding techniques. Categorical variables (STEROID, SPIDERS, and ASCITES) are labeled as 0=YES and 1=NO for a number of features in a dataset. After that all categorical values were encoded with the help of label encoding techniques (Table I). This technique converts all the variables into a numerical format. Based on clinical relevance and domain expertise, four continuous features—AGE, BILIRUBIN, SGOT, and ALBUMIN—were binned into appropriate ranges. The severity or stage of the property was then represented by encoding these ranges as ordinal levels (0, 1, 2, etc.).

TABLE I: ENCODING INFORMATION

Feature Name	Description	Binned Range	Encoded Value
AGE	Patient age in years	10–29 30-49 50-69 70 and above	0 1 2 3
BILIRUBIN	Bilirubin level (mg/dl)	0.0–0.9 1.0–2.0 >2.0	0 1 2
SGOT	SGOT (U/ml)	0–49 50–99 100–199 ≥200	0 1 2 3
ALBUMIN	Albumin (g/dl)	<3.0 3.0–3.9 ≥4.0	0 1 2

Feature Selection: Among 20 features, the features containing high amounts of missing values (PROTIME, ALK PHOSPHATE) were removed during pre-processing. The remaining 18 features which includes clinical symptoms (BILIRUBIN, SGOT, and ALBUMIN) and categorical features (FATIGUE and VARICES) were preserved because they contain vital clinical records. We used the Chi-squared (χ^2) test and XGBoost feature importance to find the most pertinent predictors. Whereas XGBoost ordered features according to how they contributed to the accuracy of classification, the χ^2 test chose features based on their statistical association with the goal. The best characteristics from both approaches were blended to form a composite set. The last 12 features that were chosen are as follows: ['SEX', 'ALBUMIN', 'BILIRUBIN', 'SPLEEN PALPABLE', 'LIVER BIG', 'AGE', 'SGOT', 'LIVER FIRM', 'ANOREXIA', 'SPIDERS', 'STEROID', 'ASCITES']. To help us illustrate these decisions and identify their connections, we created a correlation heatmap (Fig 3).

Data Balancing: The dataset contains 122 patients who have survived and 33 patients who didn't survived an oversampling technique, a hybrid technique called SMOTE-Tomek was applied to handle the dataset. Tomek Links eliminates unclear or overlapped data between classes, whereas SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic samples for the minority class. This dual approach improves the quality of the dataset while simultaneously balancing the distribution of classes

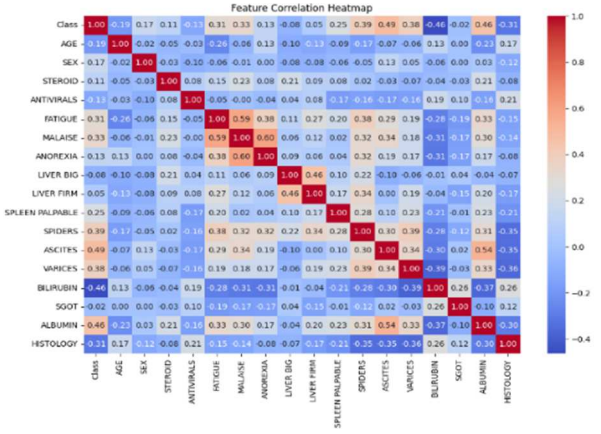


Fig. 3: Correlation between all the features

Data Splitting: Before training, the dataset used in the machine learning procedure is splits into two parts: training and the testing set. Generally, an 80:20 stratified split was used to ensure that class distribution of target variables preserved in both subsets.

C. Model Selection

In order to develop our Hepatitis B analysis model, 7 widely recognized classifiers were employed. Based on their relatively low error rate, the top-performing models are selected from the aforementioned results. The classifiers are: K-Nearest Neighbors (KNN), Naive Bayes Classifier, Support Vector Machine (SVM), Decision Tree Classifier, Logistic Regression (LR), Random Forest and XGBoost. Before putting any machine learning technique into practice, hyperparameter optimization was done. The values of the parameters utilized for each classifier are displayed in Table II.

TABLE II: HYPERPARAMETER TUNING FOR EACH MODEL

Classifier	Parameter	Value
Support Vector Machine (SVM)	C gamma	1 1
Decision Tree	max-depth min-samples leaf min-samples split	5 10 2
KNN	n-neighbors	3
Naive Bayes		
Random Forest	n-estimators max-depth	300 10
LR	C	10
XGBoost	n-estimators max-depth learning rate	200 5 0.1

A meta-learning strategy that incorporates the advantages of several classifiers is the suggested Stacked Ensemble. In this investigation, the meta-classifier was Logistic Regression, while the base learners were adjusted Random Forest, SVM,

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and XGBoost. While the meta-classifier learnt how to best aggregate the outputs of the base learners, each of them separately identified distinct patterns in the data.

IV. Result Analysis

Analyzing four important classification metrics—Accuracy, Precision, Recall, and F1-Score, Table III compares eight machine learning models: Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), Logistic Regression, Naive Bayes, Decision Tree, random Forest, XGBoost & the proposed stacked ensemble.

TABLE III: EVALUATION OF THE EXPLANAINABLE ASPECTS OF EIGHT MACHINE LEARNING MODELS

Index	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	Support Vector Machine	81.00	98.00	62.5	76.9
2	Random Forest	91.7	95.00	87.5	91.3
3	K-Nearest Neighbors	93.8	88.5	99.0	91.3
4	Logistic Regression	88.5	88.00	89.00	88.5
5	Naive Bayes	75.6	98.00	62.5	76.9
6	Decision Tree	82.9	85.00	80.00	82.4
7	XGBoost	88.00	92.00	85.00	88.00
8	Stacked Ensemble (proposed)	97.9	99.00	91.7	95.6

With a performance of 93.8% on all models in metrics, KNN outperformed the other models. While Random Forest followed closely at 91.7%. Logistic Regression (88.5%) and XGBoost (88.0%) showed competitive results, outperforming the Decision Tree classifier (82.9%). In contrast, SVM (81.2%) and Naive Bayes (75.6%) shows lower recall which indicates limitations in detecting TP cases despite high precision values. The proposed stacked ensemble model outperformed all other classifiers, achieving an accuracy of 97.9%, precision close to 99%, and an F1-score of 95.6%. This makes it very evident that stacking Random Forest, SVM, and XGBoost predictions greatly improves classification performance when compared to utilizing individual models. Confusion matrices (Fig. 4), which show the distribution of accurate and inaccurate predictions for each model, provided additional support for the evaluation. The Stacked Ensemble obtained the highest area under the curve (AUC), indicating its greater capacity to distinguish between survivors and non-survivors, according to ROC curve analysis (Fig. 5).

A. Confusion Matrix

The confusion matrix, which offers a thorough analysis of the model's classification performance, is the source of these assessment metrics.

The model's sensitivity, specificity, accuracy, recall, and F1-score may all be evaluated by examining these elements. Accuracy in Hepatitis B prediction calculate to the total number of correctly identified patient outcomes (1), whereas recall evaluates the model's capacity to identify real non-survivors (2). Precision measures the proportion of predicted non-survivors who were actually at risk (3). The F1-score strikes a compromise between recall and precision, guaranteeing dependability in medical decision-making (4). The equations are:

$$Accuracy = \frac{TP+TN}{(TP+TN+FP+FN)} \quad (1)$$

$$Sensitivity\ (Recall) = \frac{TP}{TP+FN} \quad (2)$$

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

$$F1-Score = \frac{2\times TP}{2\times TP+FP+FN} \quad (4)$$

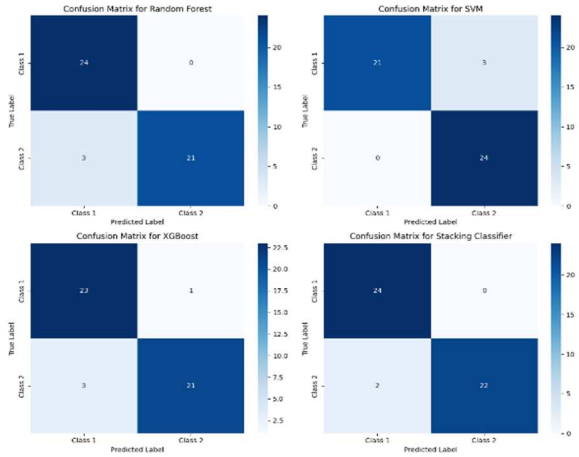


Fig. 4: Performance Analysis of Models Using Confusion Matrix.

B. ROC/AUC Curve

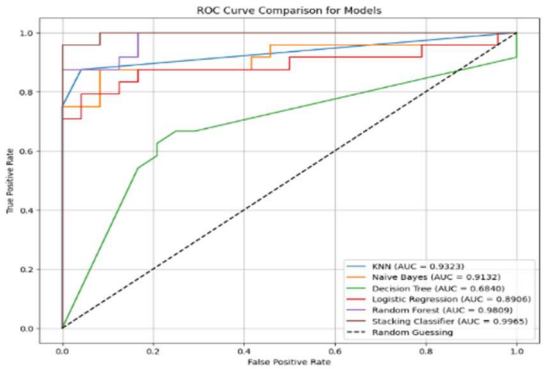


Fig. 5: Comparison of the ROC curves for eight accuracy models

The ROC curve analysis in Fig. 5 clearly illustrates how well each machine learning model separates survivors from non-survivors in the Hepatitis B dataset.

TABLE IV: COMPARATIVE ANALYSIS OF VARIOUS MACHINE LEARNING MODELS USED FOR THE CLASSIFICATION AND EARLY PREDICTION OF HEPATITIS B INFECTION

Paper	Data modality	Dataset used	Classification	Accuracy
Obaido, G. et al. [10]	Clinical data	Kaggle / Custom clinical records	AdaBoost, XGBoost,	Up to 92%
Ye, Y.-M. et al. [16]	Clinical data	Clinical patient data.	LR, RF, SVM	95%

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Ajuwon, B. I. et al. [17]	Clinical data	Nigerian clinical dataset	SVM, NB, Decision Tree	85.4%
Moulaei, K., Sharifi et al. [18]	Clinical data	Not specified	SVM	~90%
Proposed Model	***Clinical data***	UCI Repository (Hepatitis B dataset)	Stacked Ensemble (RF + SVM + XGBoost)	***97.9%***

With an accuracy of 97.9%, our model outperformed earlier published techniques, which generally ranged between 85% and 95%. Although previous studies contributed significantly to the field but many of these studies focused on particular classifiers and lacked a thorough strategy for dealing with missing data or class imbalance. By using a strong preprocessing pipeline that includes KNN imputation, SMOTE-Tomek resampling and label encoding, our approach overcomes these challenges. The model captures a greater range of decision patterns by combining Random Forest, SVM, and XGBoost in a stacked ensemble framework, which leads to more accurate and dependable projections of Hepatitis B outcomes.

V. Conclusion

The paper highlights how structured clinical data and thorough preparation may be used to effectively predict Hepatitis B outcomes using machine learning. By integrating KNN-based imputation, label encoding, and SMOTE-Tomek hybrid resampling, the dataset was effectively cleaned, balanced, and prepared for model training. Multiple classifiers were evaluated, including KNN, Naive Bayes, Decision Tree, Logistic Regression, Random Forest, SVM, and XGBoost. The suggested Stacked Ensemble model, which combines adjusted Random Forest, SVM, and XGBoost with Logistic Regression as the meta-classifier, achieved the greatest accuracy of 97.9% with good precision and F1-scores, despite the strong performance of individual models like KNN and Random Forest. These results demonstrate that ensemble-based methods provide a dependable and accurate solution for Hepatitis B prognosis when supported by strong preprocessing techniques.

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